

Mini Project Documentation: Load Balancing using AWS

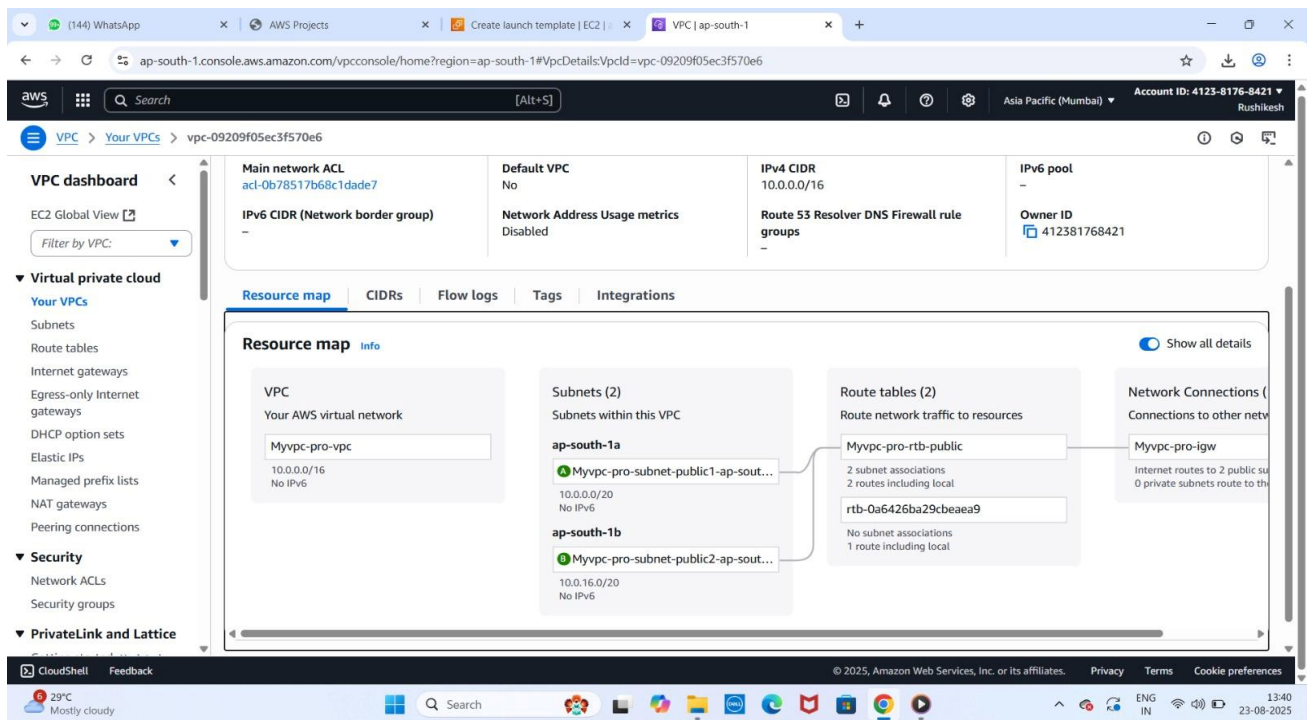
Objective

The goal of this project is to set up a highly available web application using an Application Load Balancer (ALB) in AWS. The project demonstrates how to distribute traffic across multiple web servers running in different Availability Zones (AZs) for fault tolerance and scalability.

Steps Involved

Step 1: Create a VPC with Subnets

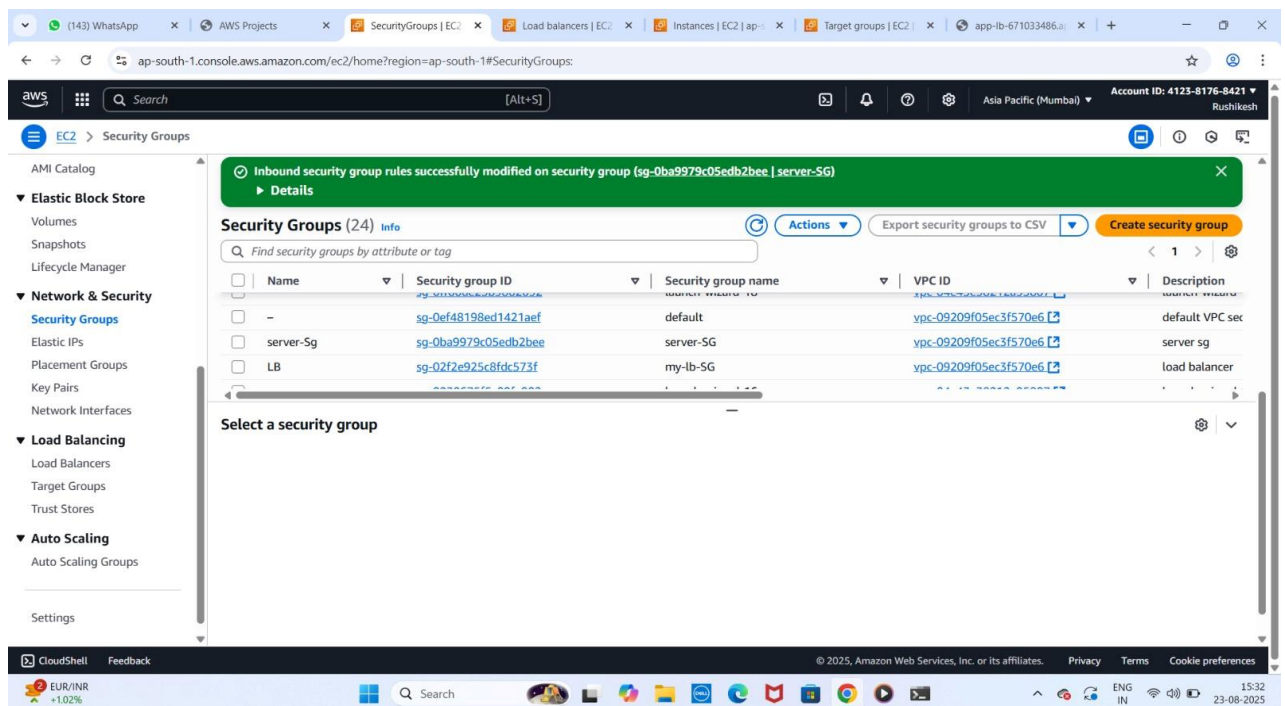
- A new **VPC** was created with **2 Availability Zones (AZs)**.
- Within the VPC, **2 public subnets** were configured, one in each AZ, to host web servers and the load balancer.



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Step 2: Configure Security Groups

- **Security Group 1 (for Load Balancer):**
 - Inbound: Allowed HTTP (Port 80) and HTTPS (Port 443) traffic from the internet.
- **Security Group 2 (for Web Servers):**
 - Inbound: Allowed only HTTP traffic from the Load Balancer's Security Group.
 - This ensures secure communication between the load balancer and backend servers.

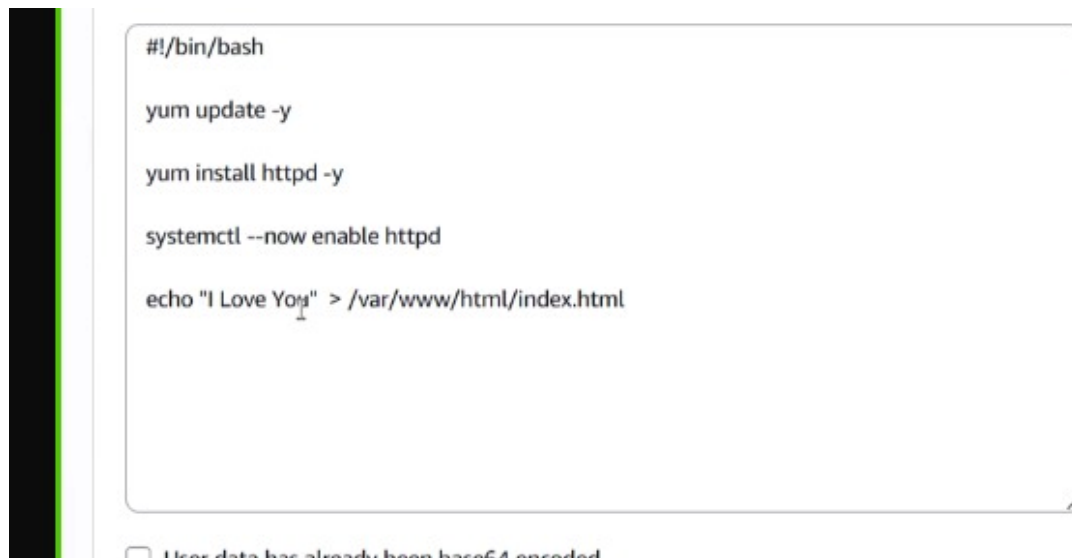


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Step 3: Create Launch Template for EC2 Instances

- A **Launch Template** was created to simplify instance provisioning.
- The template included:
 - Security group for servers.
 - Amazon Linux 2 AMI.
 - Apache HTTP server installation script.
 - A simple **index.html** file with a custom message.

User Data Script Used:



```
#!/bin/bash

yum update -y

yum install httpd -y

systemctl --now enable httpd

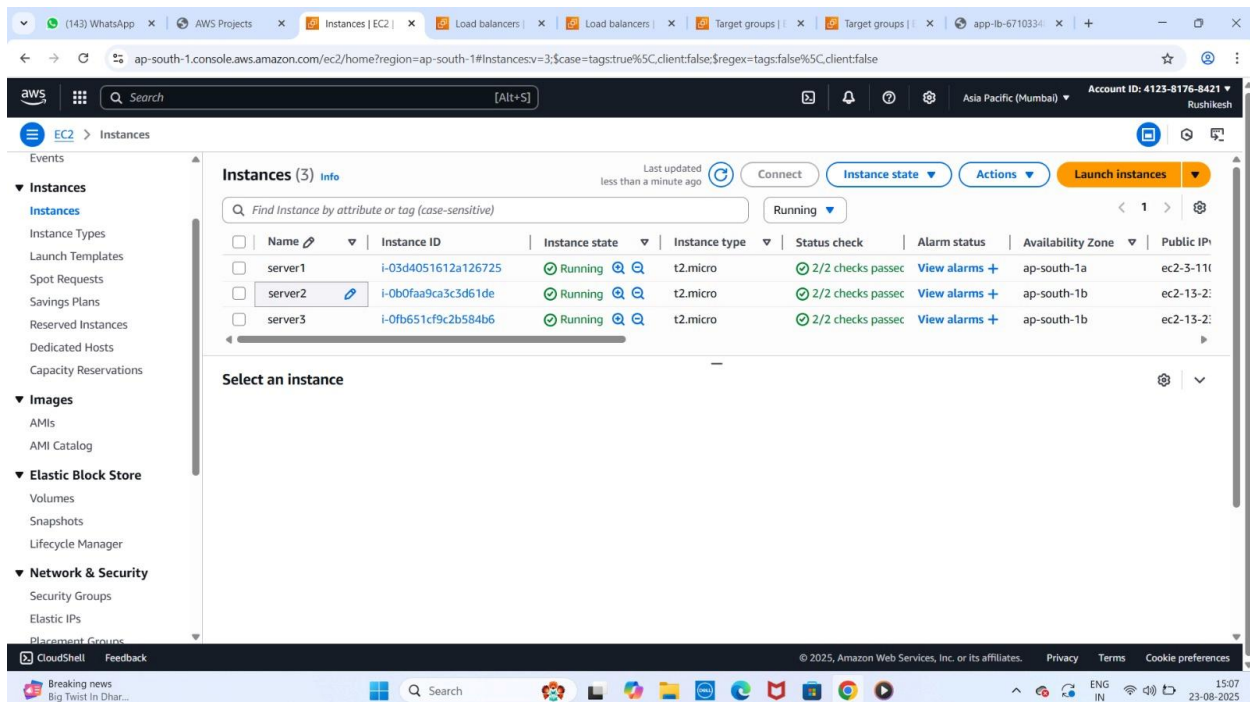
echo "I Love You" > /var/www/html/index.html
```

☐ User data has already been base64 encoded

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Step 4: Launch EC2 Instances (Web Servers)

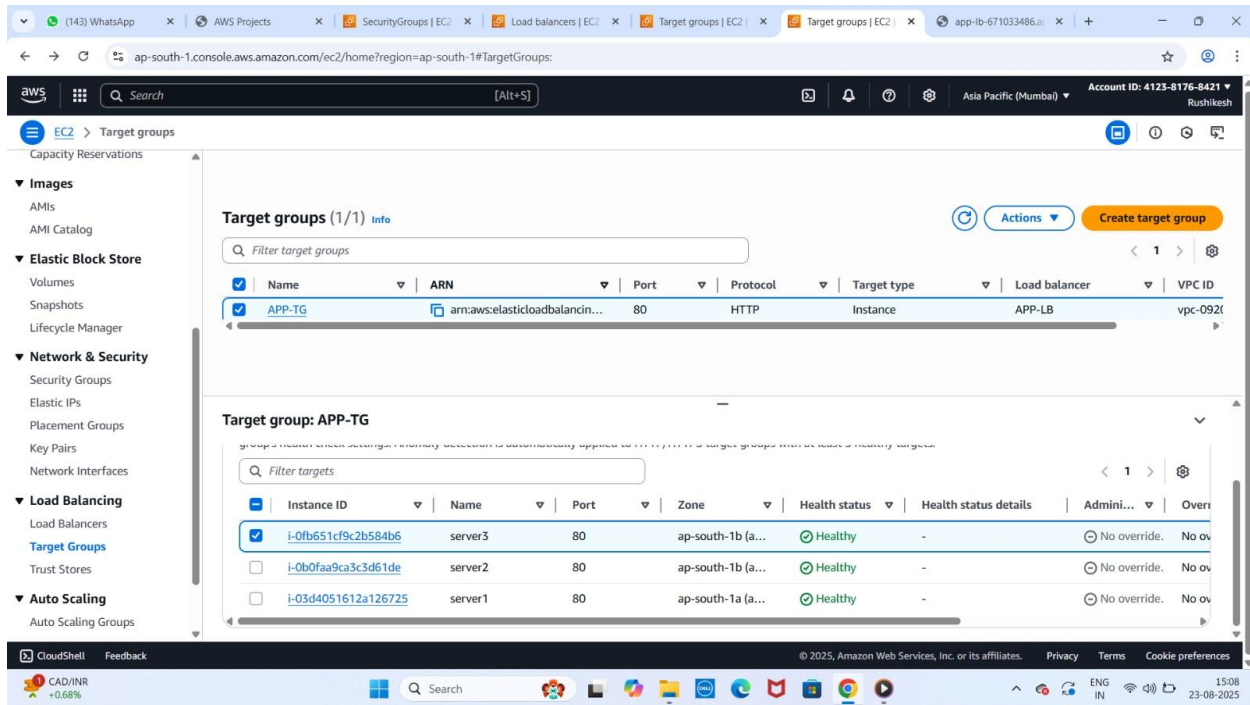
- **3 EC2 instances** were launched using the template.
- Each instance automatically installed Apache and displayed a unique message via index.html.
- All servers were associated with the **Web Server Security Group**.



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Step 5: Create Target Group and Register Instances

- A **Target Group** was created for the Application Load Balancer.
- All 3 web servers were **registered** in the target group.
- Health checks were configured to verify availability.



The screenshot displays the AWS Management Console for the 'Target groups' section. The left-hand navigation pane shows the 'Load Balancing' section expanded, with 'Target Groups' selected. The main content area shows the 'Target groups (1/1)' page. A table lists the target groups, with 'APP-TG' selected. Below this, the 'Target group: APP-TG' details are shown, including a list of registered instances.

Instance ID	Name	Port	Zone	Health status	Health status details	Admini...	Over
☒ i-0fb651cf9c2b584b6	server3	80	ap-south-1b (a...	Healthy	-	No override.	No ov
☐ i-0b0faa9ca3c3d61de	server2	80	ap-south-1b (a...	Healthy	-	No override.	No ov
☐ i-03d4051612a126725	server1	80	ap-south-1a (a...	Healthy	-	No override.	No ov

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Step 6: Create Application Load Balancer (ALB)

- An **Application Load Balancer** was created in the VPC.
- Configured to use the **Load Balancer Security Group**.
- Listener setup on Port 80 (HTTP).
- Target group attached for traffic forwarding.
- Health check interval set to **5 seconds** for fast detection of unhealthy servers.

The screenshot displays the AWS Management Console for an Application Load Balancer named 'APP-LB'. The console shows the following configuration details:

- Listeners (1):** A single listener is configured for HTTP on Port 80.
- Rules (1):** A single rule is configured with 'Priority default' and 'Forward to target group'.
- Target groups (1):** A single target group is configured for 'Instance, HTTP' and 'APP-TG'. It shows 3 targets and 3 healthy instances.
- Targets (3):** Three targets are listed, each with a unique ID and Port 80. All three targets are marked as 'Healthy'.

The console also shows the 'Load balancers (1/1)' table with the following columns: Name, VPC ID, Availability Zones, Security groups, DNS name, ARN, and Date created. The table shows one entry for 'APP-LB' with VPC ID 'vpc-09209f05ec3f570e...', 2 Availability Zones, Security group 'sg-02f2e925c8fcd573f', DNS name 'APP-LB-671033486.ap-sout...', ARN, and Date created 'August 23, 2025, 14:31 (UTC+0...)'. The console also shows the 'Load balancer: APP-LB' section with a diagram of the configuration.

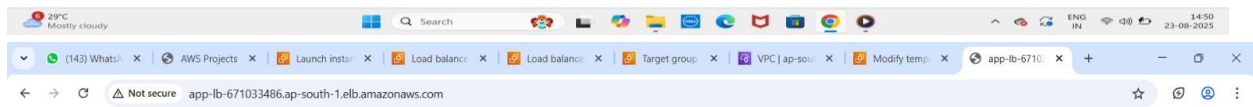
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Step 7: Access Application via Load Balancer DNS

- The DNS name of the ALB was copied.
- Accessing the DNS in the browser distributed traffic across all 3 servers.
- Refreshing the page showed responses from different servers, proving load balancing.



I Love You



I Hate You



Conclusion

This mini project successfully demonstrated:

- Hosting multiple web servers across different Availability Zones.
- Restricting access with proper security group configuration.
- Using an Application Load Balancer to distribute incoming traffic.
- High availability and fault tolerance through health checks and multiple instances.